

Ion Thruster and VLEO Findings

Consolidated Research Note

18 March 2026

1 Scope

This note consolidates the earlier research thread into a single cited reference. It now covers five questions:

1. What is the maximum thrust produced by ion thrusters?
2. What is the lowest altitude at which a spacecraft has orbited Earth?
3. What is the minimum specific impulse, I_{sp} , of an ion thruster?
4. What is a typical specific impulse, I_{sp} , for an ion thruster?
5. What is the maximum specific impulse, I_{sp} , documented for an ion thruster?

Two terminology caveats matter for interpreting the answers:

- “Ion thruster” is used loosely in public discussions. In narrow usage it usually means gridded ion engines; in broader usage it may include Hall thrusters and electrospray or colloid ion thrusters.
- “Most powerful in space” can mean either the most powerful *single engine* or the largest *combined spacecraft propulsion system*. Those are not always the same record.

2 Maximum Thrust from Ion Thrusters

Table 1 summarizes the best-supported values found in non-Wikipedia sources.

The main conclusions are:

- The largest thrust found for an ion thruster in testing is the X3 Hall thruster at **5.4 N** [1, 2].
- For a *single flown ion engine*, the strongest documented capability found in the reviewed sources is **NEXT-C at up to 235 mN** [3, 4].

Table 1: Documented ion-thruster thrust figures

Case	Thruster or system	Status	Thrust	Notes
Ground-test maximum	X3 nested-channel thruster	Hall Ground test	5.4 N	Demonstrated at 102 kW with 1800–2650 s specific impulse [1, 2]
Single ion engine used in space	NEXT-C	Flown	Up to 235 mN	NASA Glenn fact sheet gives a 25–235 mN operating range; Guinness lists NEXT-C as the most powerful ion engine used in space [3, 4]
Combined documented spacecraft ion-propulsion thrust	BepiColombo solar-electric system	MTM propulsion Flown	290 mN total	ESA states the Mercury Transfer Module provides up to 290 mN total thrust; QinetiQ T6 operating points are 75–145 mN per thruster with up to 4300 s specific impulse [5, 6, 7]

- For a *spacecraft system*, ESA documents **up to 290 mN total** for the BepiColombo Mercury Transfer Module electric-propulsion system [5, 6].

So, if the question is about a single flown ion engine, the best documented answer is **235 mN**. If the question is about the total thrust available from a flown spacecraft ion-propulsion system, the best documented answer found here is **290 mN**. If the question includes ground-tested hardware, the answer rises to **5.4 N**.

3 Lowest Altitude Spacecraft Orbit

The best-supported answer is JAXA’s **TSUBAME** (SLATS), which set the record for the lowest altitude by an Earth-observation satellite in orbit at **167.4 km** [8, 9].

JAXA’s public material identifies TSUBAME as the record-holder at 167.4 km [8]. Guinness World Records gives the same altitude and dates the mission from 23 December 2017 to 1 October 2019 [9]. JAXA also explains why this regime is difficult: at 200–300 km altitude the atmospheric drag and atomic-oxygen environment are about 1000 times greater than those experienced by most Earth-observation satellites, which is why TSUBAME required active orbit maintenance with an ion engine and why the mission was treated as a technology demonstrator [10].

Therefore, the lowest orbital altitude found in the reviewed sources is **167.4 km**.

4 Specific Impulse of Ion Thrusters

There is no single universal minimum, typical, or maximum without defining the thruster class and whether advanced concepts are included.

Specific impulse is related to effective exhaust velocity by

$$I_{sp} = \frac{v_e}{g_0}. \quad (1)$$

4.1 Minimum values

For *conventional* gridded ion engines, the reviewed NASA sources place operating ranges well above chemical-propulsion values:

- NASA’s low-power ion-propulsion work for small spacecraft reports approximately **2000 to 3000 s** [11].
- NASA reporting on high-power ion thrusters gives approximately **3000 to 5000 s** [12].

For *electrospray or colloid ion thrusters*, lower values are documented:

- An MIT open-access journal paper reports a miniaturized ionic-liquid electrospray thruster at **about 760 s**, including all losses, with 11–12.5 μN thrust [13].
- MIT’s S-iEPS CubeSat system is reported at **1200 s** and 100 μN with less than 1.5 W input power [14].
- Busek lists its BET-MAX electrospray system at **850 to 2300 s** [15].
- An IEPC electrospray performance paper reports a much broader capability range of **100 to 10000 s** [16].

The practical conclusion on the minimum is therefore scope-dependent:

- If “ion thruster” means a conventional primary-propulsion ion engine, the minimum practical range in the reviewed sources is roughly **2000 s**.
- If electrospray and colloid ion thrusters are included, the lowest clearly documented experimental value found in accessible non-Wikipedia sources is **about 760 s** [13].
- The broader literature reviewed here indicates that electrospray devices can extend lower still, to around **100 s** in some operating regimes [16], but that value should be interpreted as technology- and mode-dependent rather than as a universal minimum for all ion thrusters.

4.2 Typical values

For conventional gridded ion engines, the reviewed non-Wikipedia sources cluster in the low-thousands of seconds:

- NASA’s low-power ion-propulsion work for small spacecraft reports approximately **2000 to 3000 s** [11].
- NASA reporting on high-power ion thrusters gives approximately **3000 to 5000 s** [12].
- For CubeSat-class hardware, Busek’s BIT-3 RF ion thruster product page reports **up to 2150 s**, and the associated IEPC paper describes a 60 W design point of **3200 s** [17, 18].

A practical engineering answer is therefore that a *typical conventional ion thruster* operates at about **2000 to 4000 s**, with smaller or lower-power systems nearer the lower end and larger or higher-power systems extending toward about 5000 s in the reviewed NASA literature.

4.3 Maximum values

For the maximum specific impulse, it is important to distinguish between present conventional hardware and advanced concepts:

- ESA’s DS4G overview states that present gridded ion thrusters are practically limited to **less than 10000 s** because conventional grid systems must keep beam potential around 5 kV to avoid beam divergence, impingement, and sputtering [19].
- The same ESA source gives a projected DS4G operating point of **19300 s** specific impulse, **2.5 N** thrust, and **250 kW** power for a 20 cm xenon thruster using a dual-stage 4-grid arrangement [19].

The best-supported non-Wikipedia maximum found in this review is therefore **19300 s** for the advanced ESA DS4G concept. For currently conventional ion thrusters in the reviewed NASA sources, a more practical upper-end figure is about **5000 s**.

5 Bottom-Line Answers

Using the strongest non-Wikipedia sources gathered in this review:

- **Maximum demonstrated ion-thruster thrust (ground test):** 5.4 N for the X3 Hall thruster [1, 2].
- **Most powerful single ion engine used in space:** NEXT-C, with a documented capability of up to 235 mN [3, 4].
- **Largest documented combined spacecraft ion-propulsion thrust in the reviewed sources:** BepiColombo MTM, up to 290 mN total [5, 6].

- **Lowest spacecraft orbital altitude:** TSUBAME (SLATS), 167.4 km [8, 9].
- **Minimum ion-thruster I_{sp} :** about 760 s in a documented MIT electrospray microthruster study, or about 2000 s if restricted to conventional gridded ion engines [13, 11].
- **Typical conventional ion-thruster I_{sp} :** about 2000 to 4000 s, with the reviewed NASA range extending to about 5000 s depending on power class [11, 12, 17, 18].
- **Maximum ion-thruster I_{sp} found in the reviewed non-Wikipedia sources:** 19300 s for ESA’s advanced DS4G concept; ESA describes present conventional gridded ion thrusters as practically limited to less than 10000 s [19].

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